Miniproject 2: Genetic oscillator

Scientific Computing, Bridging Course, 2023

Reproducibility is an important principle underpinning the scientific method. The results obtained in e.g. an experiment should be achieved again when the study is replicated. In this mini-project you will reproduce parts of the results in a research article[1]. The article has been important in the field of systems biology, and a deterministic and a stochastic model are studied, both describing a "circadian clock". You won't (probably) be able to understand everything in it, as it requires special knowledge in biology, but you will be able to understand enough for this miniproject.

Background

Our environment contains many cyclic processes, such as the diurnal rhythm of light and darkness. Many organisms use molecular mechanisms known as "circadian clocks", to adapt to such changes. It simultaneously promotes the expression of a negative element, which in turn represses the positive element. The cycle completes itself upon degradation of the negative element and re-expression of the positive element. In the article[1], a circadian clock model containing two proteins is used: an *activator* protein A and a *repressor* protein B. The corresponding activator and repressor genes are denoted by D_A and D_R, respectively. (D'_A and D'_R denote the same genes with A and R, respectively, bound to them.) See the article[1] for a more detailed explanation.

Preparations

You will use Python to solve the problems in the assignments below. For part (a), you will require the solver `solve_ivp` from the Python library `scipy.integrate`. For parts (b) and (c) you have two options to choose from:

Option 1: Use the Python library `Gillespy2`. This library is the same one used in the last part of Lab 4, and it is likely most convenient to work within the `stochSS` environment.

Option 2: Alternatively, you can employ the SSA code provided in the Studium, accessible via `Modules => Lecture Notes and Presentations => Lecture 10`. You would need to adapt the propensity functions, parameters, and the stoichiometry matrix according to the new model.

In both options, you must include the code in your mini-project report. To write your report,

[1] *José M. G. Vilar, Hao Yuan Kueh, Naama Barkai, Stanislas Leibler : Mechanisms of noise-resistance in genetic oscillators*, PNAS April 30, 2002 vol. 99 no. 9 page 5988-5992. The article is available in Studium.

you can utilize `stochSS` for both options. Begin by creating a new folder named for example `mini-project` and in that folder open a new Jupyter notebook (`File => New Notebook => Python3`). You add cells for code or text (LaTeX) by clicking on `Insert` or `+` tab. Switch between `code` and `markdown` modes for writing code and text, respectively. You can also use the `Heading` mode for crafting titles and section headings. Upon completing a cell, click `run`, and then add a new cell as needed. You can copy code snippets from labs and lectures, making modifications as required. Finally, download your report as a PDF file via `File => Download as => PDF via LaTeX`.

If you do not want to work with `StochSS`, you can work on a Python environment in your own Laptop that you are familiar with. In this case, if you opt for option 1, you will need to install `Gillespy2` in your environment, which may require some effort. For option 2 just copy and paste the code and modify it for the new model. Lastly, you must write your report in LaTeX or Word format (include code snippets) and then convert it into a PDF file. The choice of your preferred workflow is up to you.

Assignment

a) (Deterministic) Reproduce the result in Fig 2a, and Fig 2b in the article by solving the ODE model given in page 1 (Eq. [1]) to see how both proteins A (activator) and R (repressor) vary over 400 hours. The model parameters and initial values to be used are given in the caption of Fig. 1. Use build in function `solve_ivp` and choose a suitable method to solve the problem, and provide a rationale for your choice.

   Evaluate the performance of the three built-in ODE solvers, `RK45`, `BDF`, and `Radua` in terms of the time step sizes they employ during the simulation. This can be accomplished by generating plots that illustrate the time step sizes or reporting the number of steps used in each methods. It suffices to observe this behavior over a two-day period, equivalent to 48 hours. Explain the results (the broad results, not the details of the graphs). Which solver appears to be more suitable in this particular case and why?

b) (Stochastic) The stochastic model used in the article (it is not explicitly in the text) is the following: (16 reactions)

$$
\begin{array}{rcl}
A + R & \xrightarrow{\gamma_C} & C \\
A & \xrightarrow{\delta_A} & \emptyset \\
C & \xrightarrow{\delta_A} & R \\
R & \xrightarrow{\delta_R} & \emptyset \\
D_A + A & \xrightarrow{\gamma_A} & D'_A \\
D_R + A & \xrightarrow{\gamma_R} & D'_R
\end{array}
\qquad
\begin{array}{rcl}
D'_A & \xrightarrow{\theta_A} & A + D_A \\
D_A & \xrightarrow{\alpha_A} & D_A + M_A \\
D'_A & \xrightarrow{\alpha'_A} & D'_A + M_A \\
M_A & \xrightarrow{\delta_{MA}} & \emptyset \\
M_A & \xrightarrow{\beta_A} & A + M_A
\end{array}
\qquad
\begin{array}{rcl}
D'_R & \xrightarrow{\theta_R} & A + D_R \\
D_R & \xrightarrow{\alpha_R} & D_R + M_R \\
D'_R & \xrightarrow{\alpha'_R} & D'_r + M_R \\
M_R & \xrightarrow{\delta_{MR}} & \emptyset \\
M_R & \xrightarrow{\beta_R} & M_R + R
\end{array}
$$

   This is a discrete Markov process with a continuous time parameter. Reproduce the results in Fig 2c and Fig 2d, i.e. simulate the circadian rhythm using the stochastic numerical method SSA and study how the proteins A and R vary during 400 hours. Use

the same parameter values as in part (a).

What happens when you run the simulation several times? Does the solution agree with the ODE-solution in part (a), and does it exactly agree with the solution in Fig 2c and Fig 2d? If not, why?

c) The authors of the article point out that the presence of random white noise in the stochastic model sometimes can lead to qualitative differences in comparison with the deterministic model. One illustration of this is shown in Fig. 5, and your task here is to reproduce the result in Fig 5. What is the difference between the deterministic and the stochastic solution in this particular case?

d) Contemplate and discuss the choice of mathematical models, i.e. a deterministic vs. stochastic model. When do you think the deterministic model is a better description of reality, and when is a stochastic model to prefer? Use the results above and also (part of) the article. For example, you can read on page 1: "The preceding equations [i.e. the ODE equations] would be strictly valid in a well stirred macroscopic reactor. At the cellular level, a more realistic approach has to consider the intrinsic stochasticity of chemical reactions". Also, read *Conclusions* at the end of the article.